

CS & IT ENGINEERING



Data Structure & Programming

Trees and Graphs

Lecture No.- 07



By- Satya sir

Recap of Previous Lecture



- Heap Orderings
- Time Complexity of Heap
- Time Complexity of BST
- Need to balance BST $O(n) \rightarrow O(\log_2^n)$
- Balancing : Balance Factor $\{-1, 0, +1\}$
- Balance Factor $= H(L) - H(R)$

Topics to be Covered



Rotations

- LL Rotation
- RR Rotation
- LR Rotation
- RL Rotation



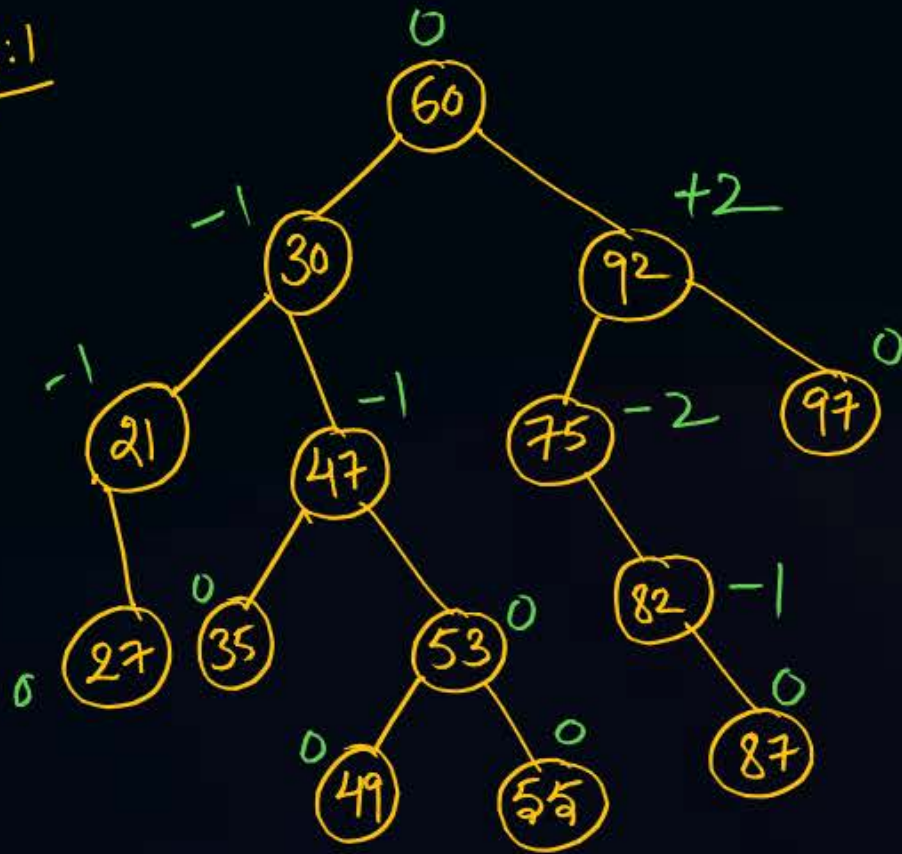


Topic : AVL Trees - 2

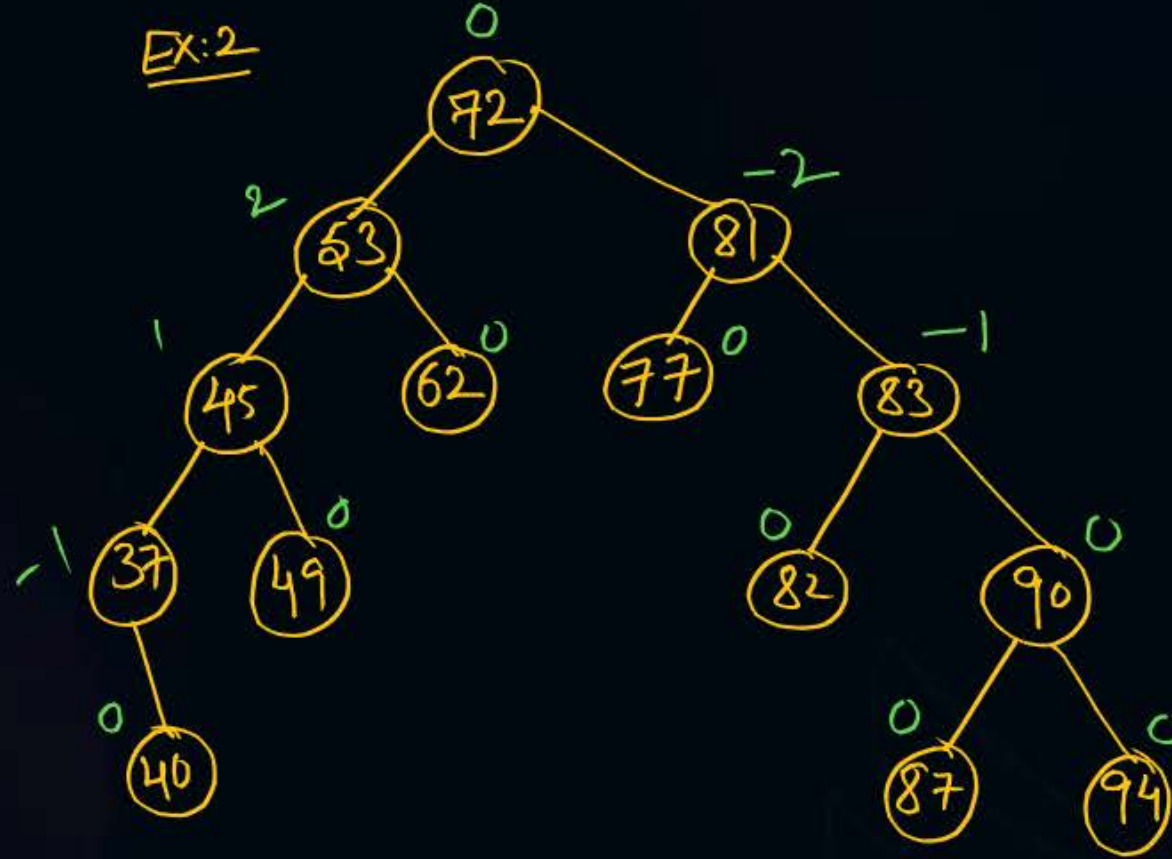


Calculate Balance Factors

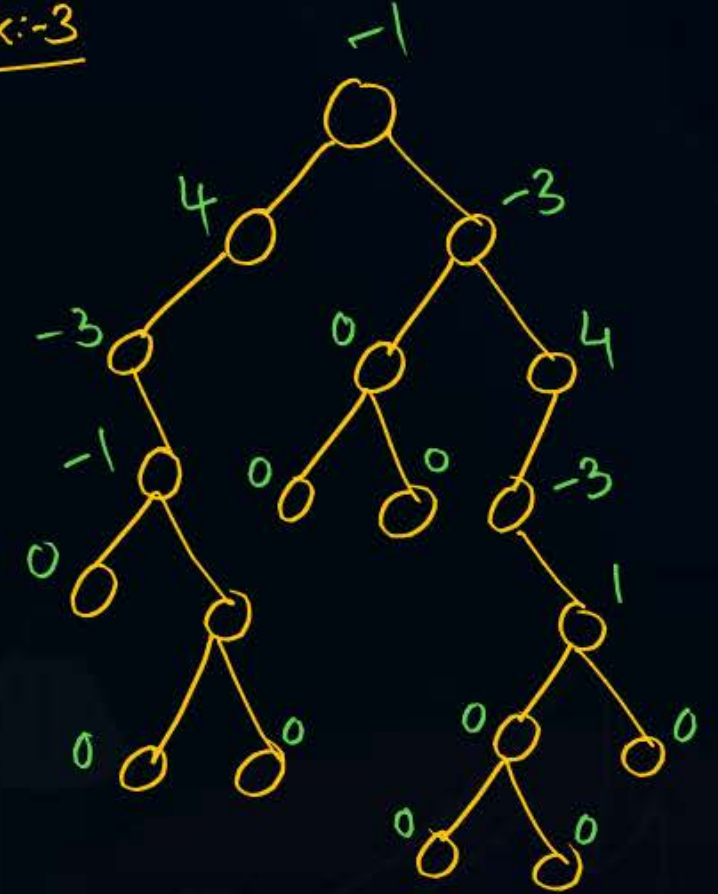
Ex:1



Ex:2



Ex:-3

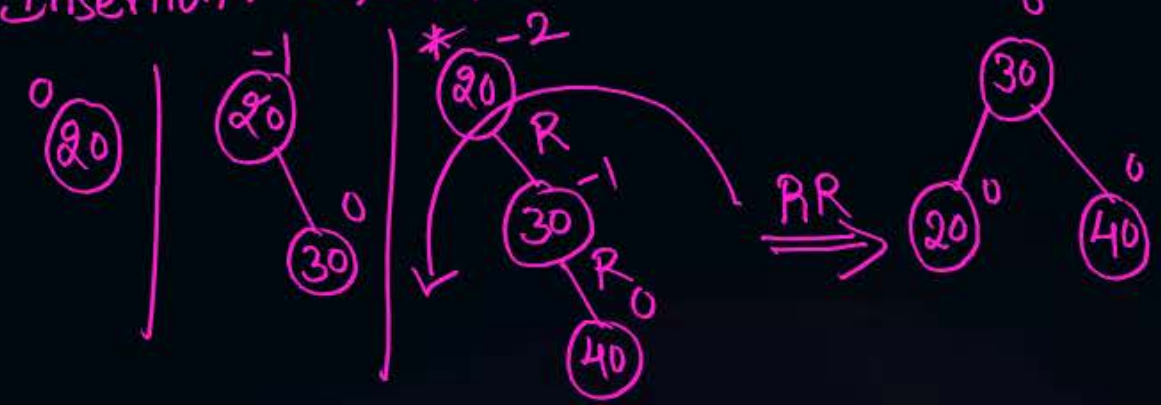




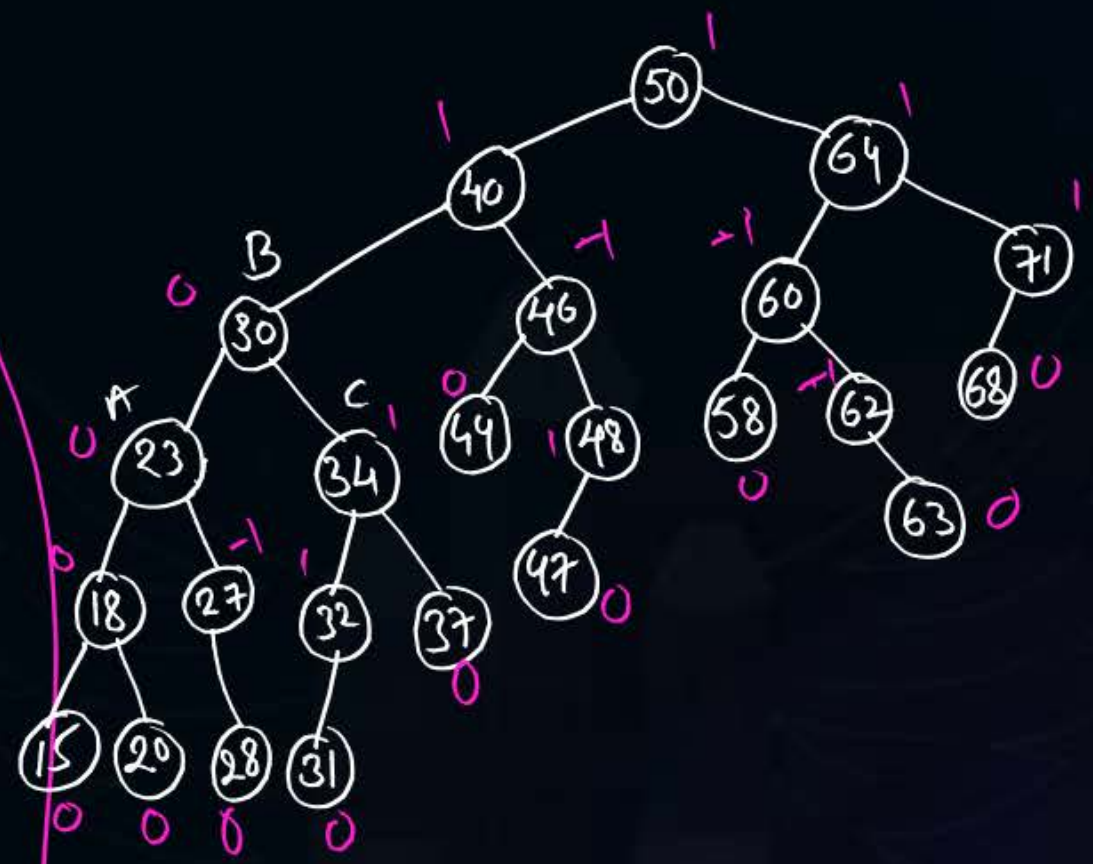
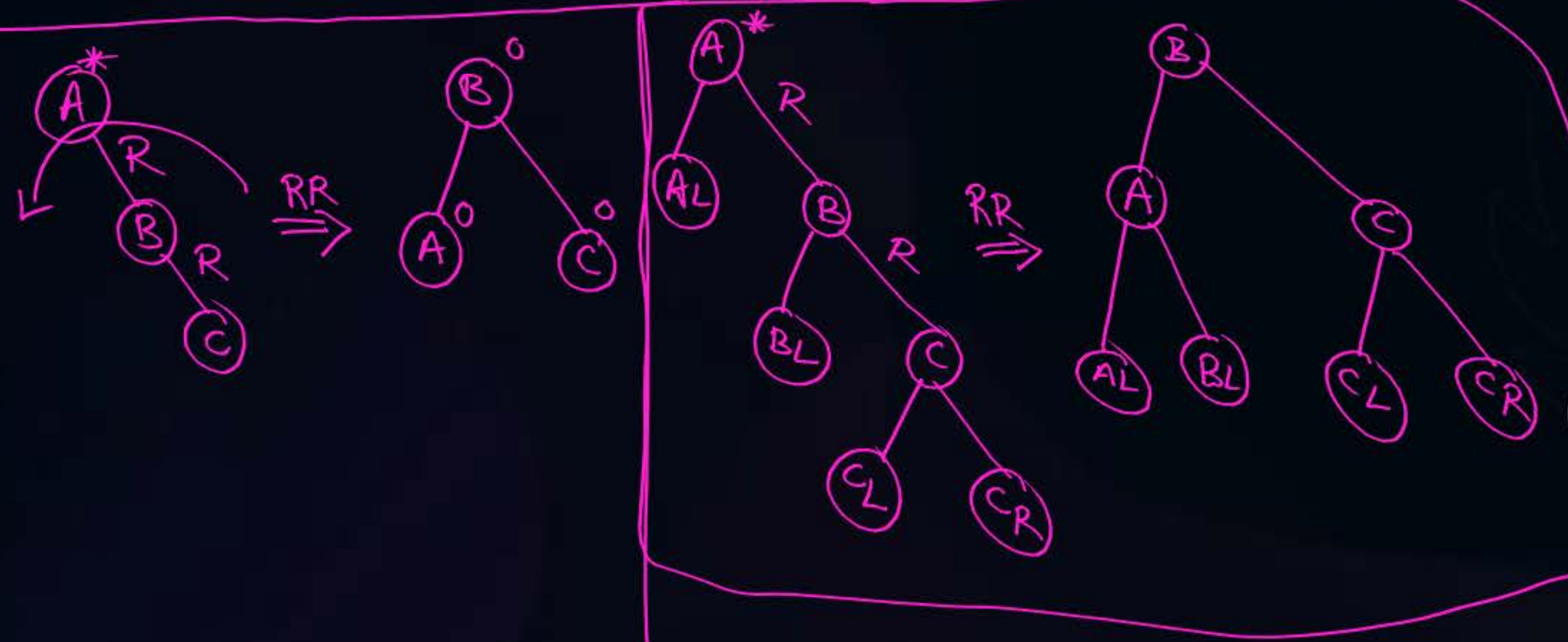
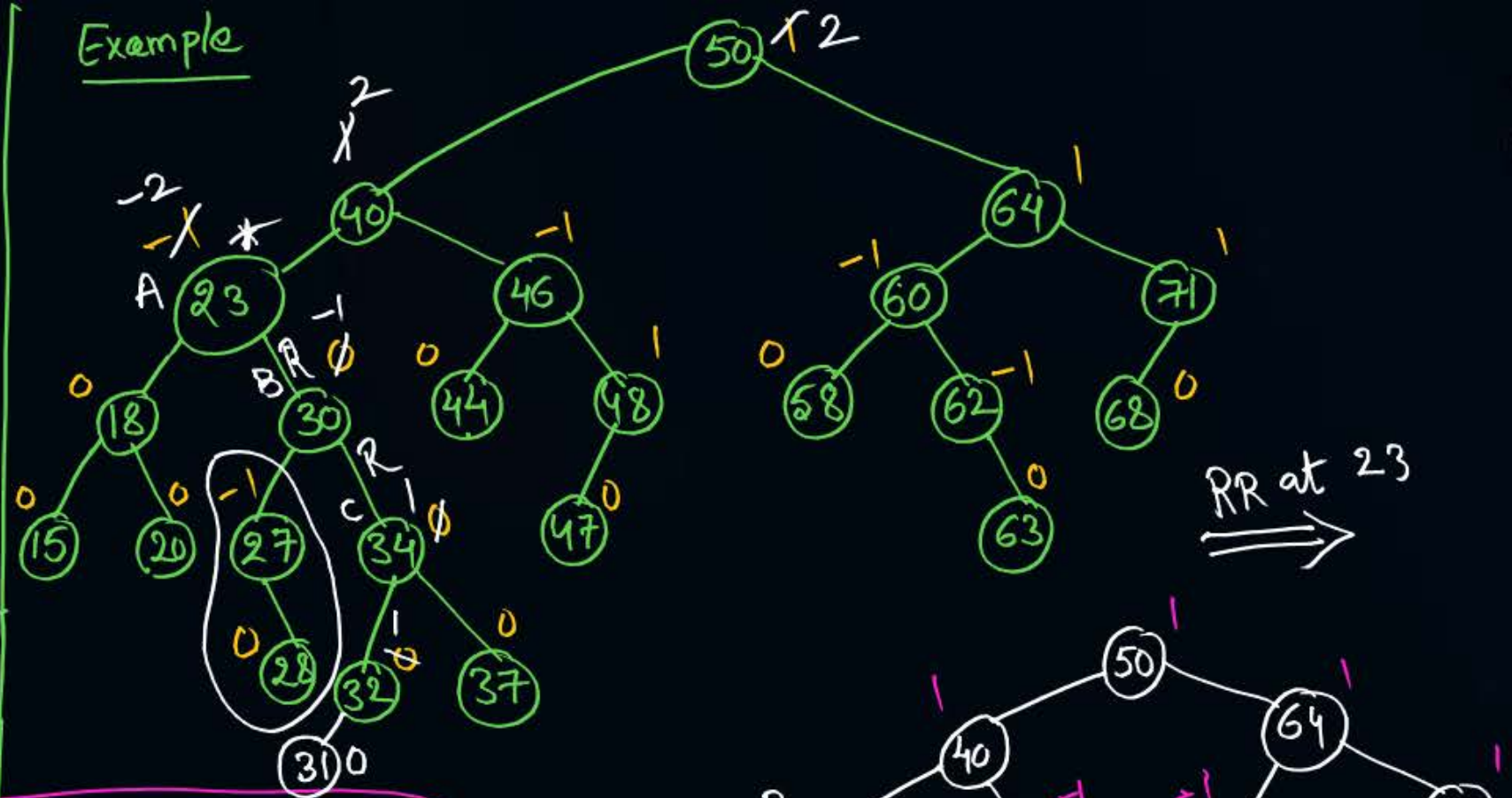
Topic : AVL Trees - 2

RR Rotation

Insertion: 20, 30, 40



Example

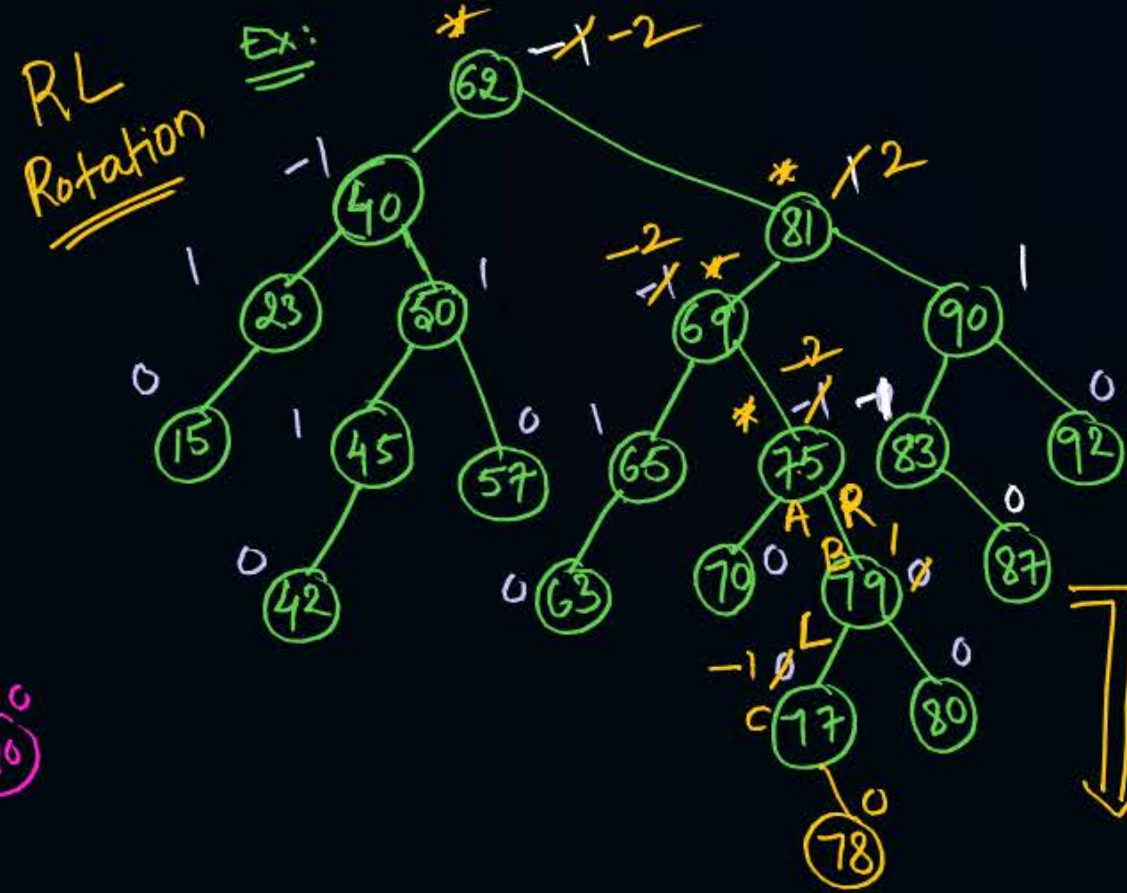
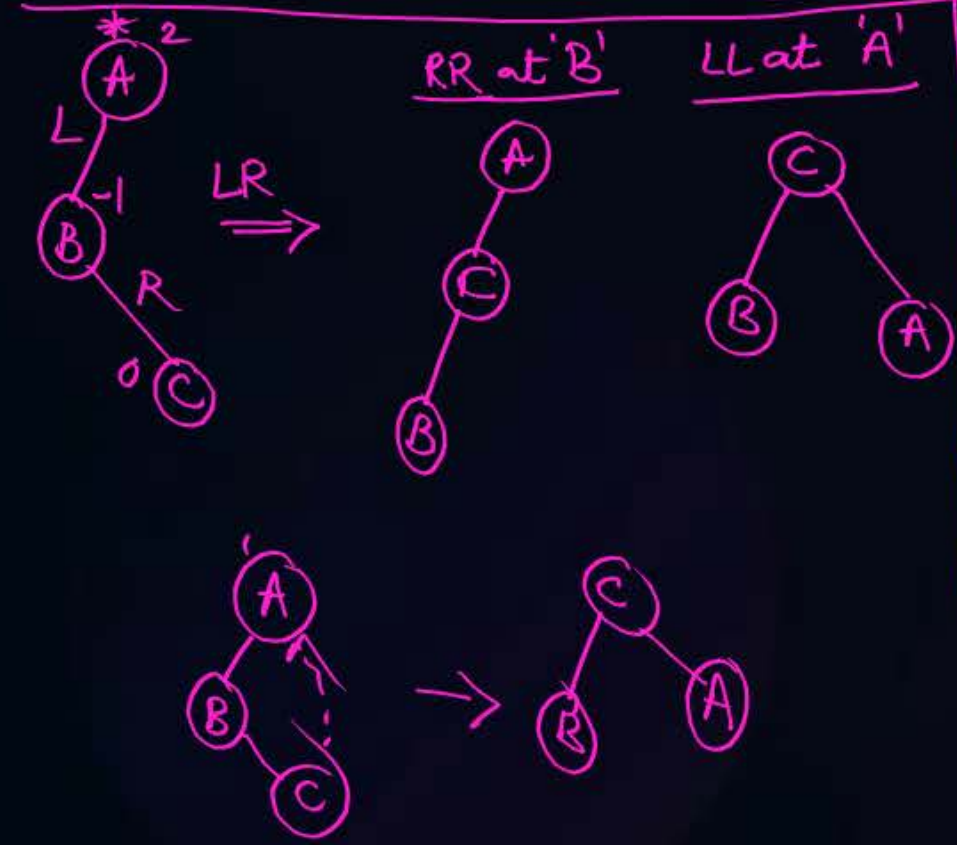
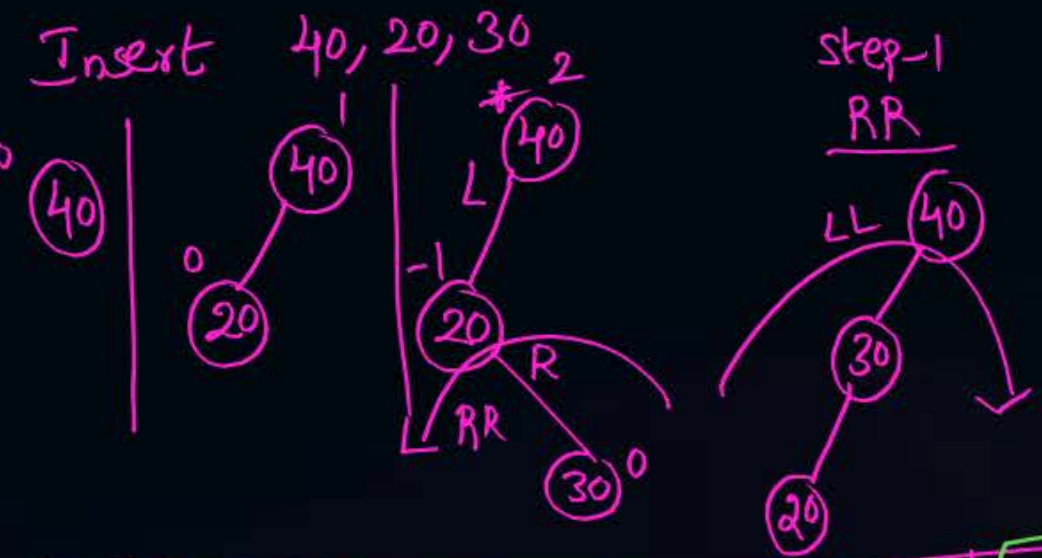




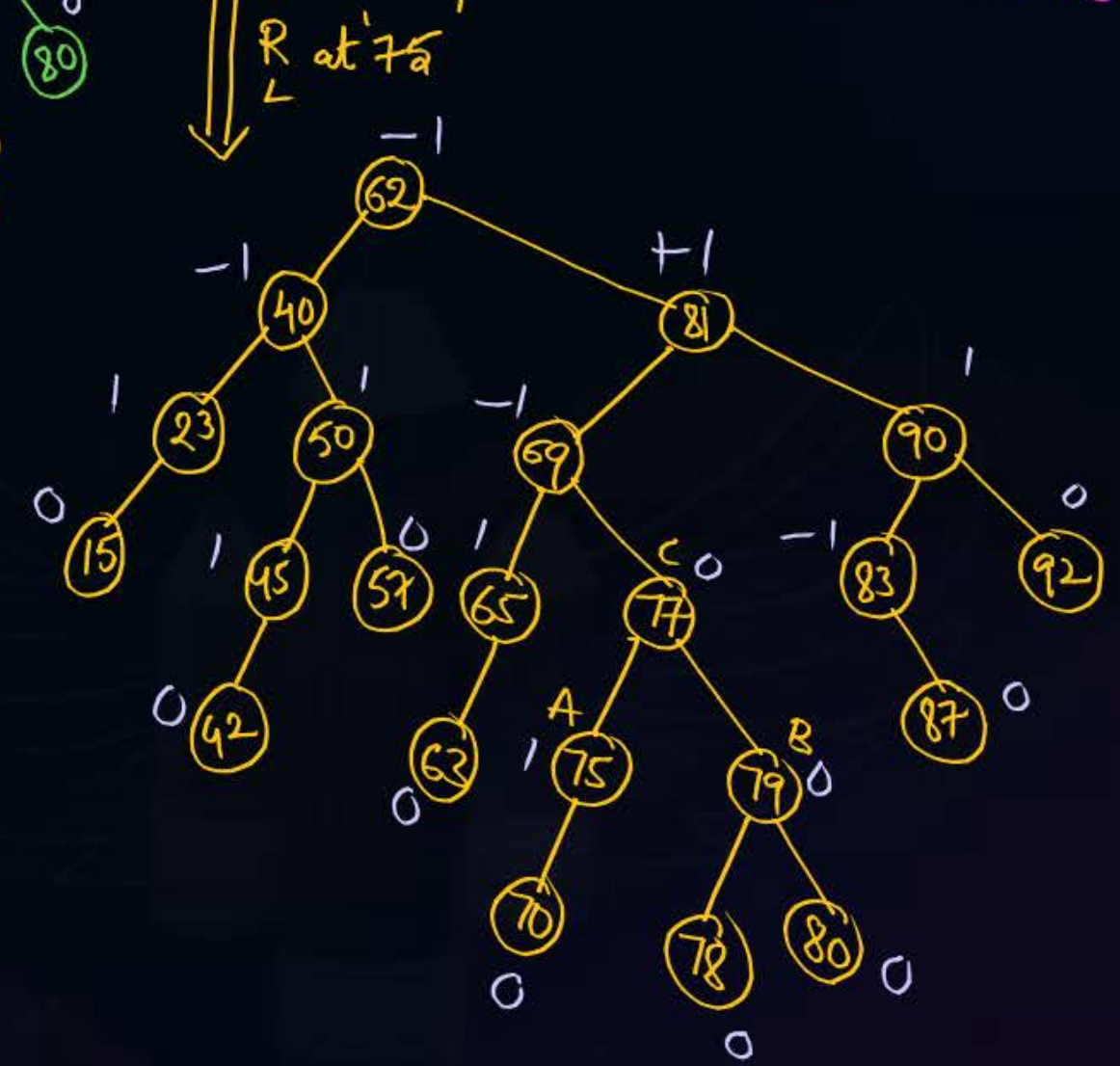
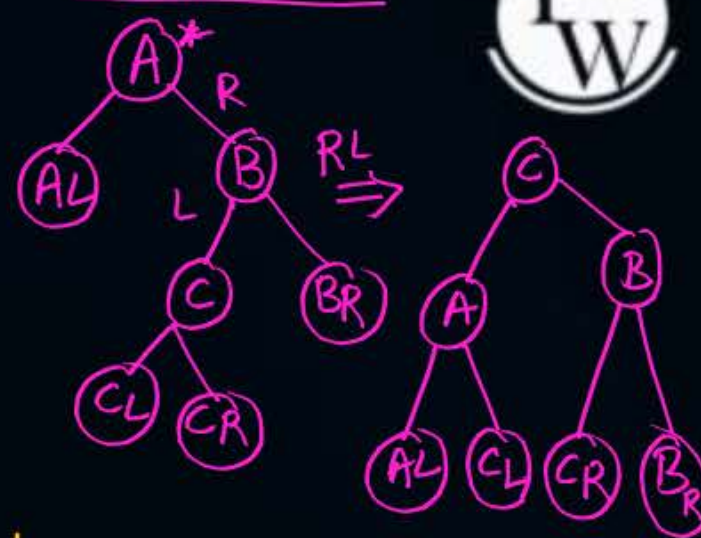
Topic : AVL Trees - 2



LR Rotation [RR + LL]



RL Rotation

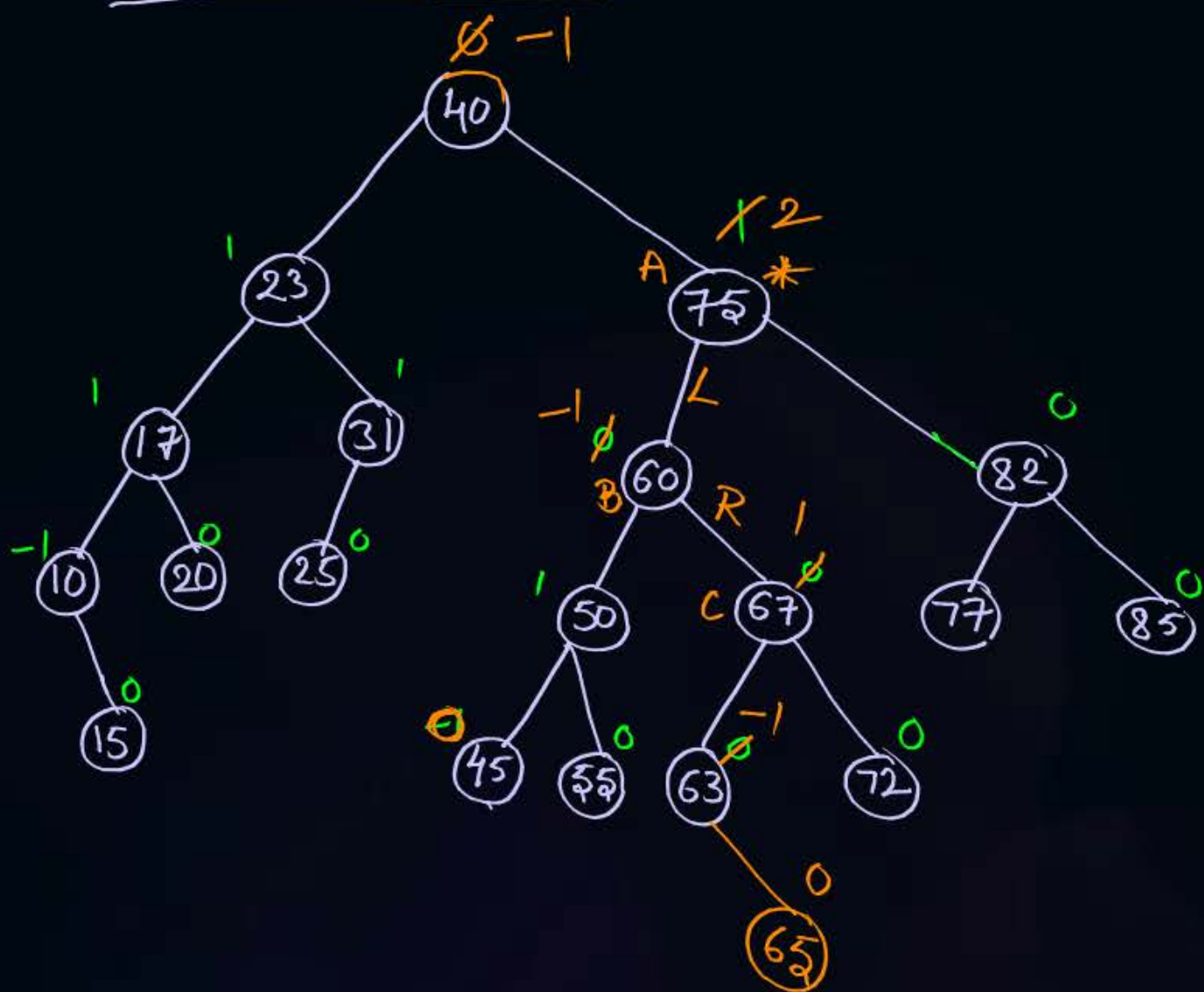




Topic : AVL Trees - 2



LR Rotation Example



LR at 75



Balanced BST



Topic : AVL Trees - 2



H/w

Construct an AVL Tree, by inserting nodes in the order: 41, 72, 65, 32, 23, 10, 5, 58, 91, 97, 99, 85, 70, 100.



2 mins Summary



Rotations

-LL

-RR

-LR

-RL

To be contd...



THANK - YOU